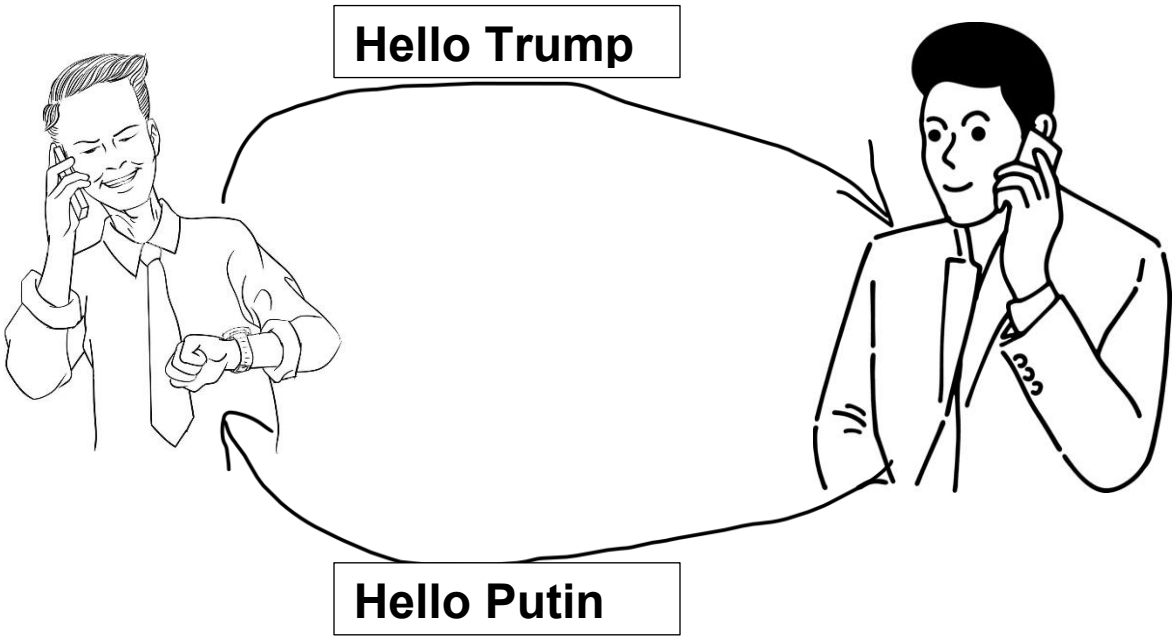
Computer Networking

Addressing

Class Material



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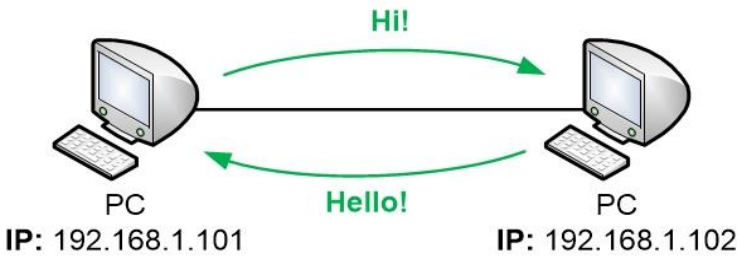
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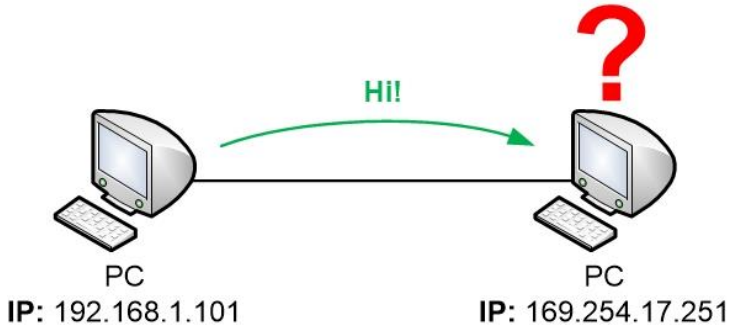
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1| What is Addressing

Addresses can be said to consist of multiple parts, similar to how full street addresses consist of multiple parts. A street address can consist of the increasingly specific parts Country, City, Postal Code, Street and House Number





2| Type of Addressing

a. **IP Address**
b. **MAC Address**
c. **URL/Hostname**

IP Address

The IP address refers to a unique address that identifies devices on the local network or internet. It is an identifier that allows information to be sent among devices on the network. An IP address is a network address of the computer that allows communication with other devices over the internet. The IP address can either be static or dynamic.



MAC Address

To communicate or transfer data from one computer to another, we need an address. In computer networks, various types of addresses are introduced; each works at a different layer. A MAC Address which stands for Media Access Control Address, is a physical address that works at the Data Link Layer.



9C:35:5B:5F:4C:D7

Difference Between MAC Address and Ip Address

Parameter	MAC Address	IP Address
Full form	Media Access Control Address	Internet Protocol Address.
Size	Six-byte hexadecimal address	Four-byte (IPv4) and sixteen-byte (IPv6)
Provider	NIC Card’s Manufacturer	Internet Service Provider
Purpose	Identifying the device	Identifying the device connection on network
Modification	Not possible	Possible
Layer of Operation	Data link layer	Network layer

URL/HOSTNAME

A URL or Uniform Resource Locator is a Unique identifier that is contained by all the resources available on the internet. It can help to locate a particular resource due to its uniqueness. It is also known as the web address. A URL consists of different parts like protocol, domain name, etc.



Types of IP Version

Currently, there are 2 versions of IP addresses in use IPV4 and IPV6

- IPV4 It is the first version of the Internet Protocol address. Size of IPv4 is a 32-bit number. In this version, Internet Protocol Security for Network security is optional. It has 4,294,967,296 addresses, but we are still seeing a shortage in network addresses as the use of network and virtual devices is increasing rapidly. Here are four numbers (called octets), and each one ranges from 0 to 255. Example: 192.168.1.5 is valid, but 192.300.1.5 is invalid because 300 exceeds 255.

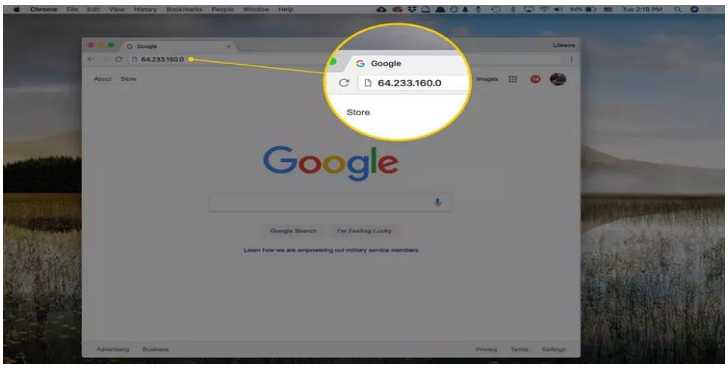
- IPV6 (Internet Protocol Version 6): It is the recent version of the Internet Protocol address. The address size of IPv6 is 128-bit number. In this Internet Protocol Security (IPSec) with respect to network security is mandatory. It allows 3.4 x 10^38 unique IP addresses which seems to be more than sufficient to support trillions of internet devices present now or coming in future.



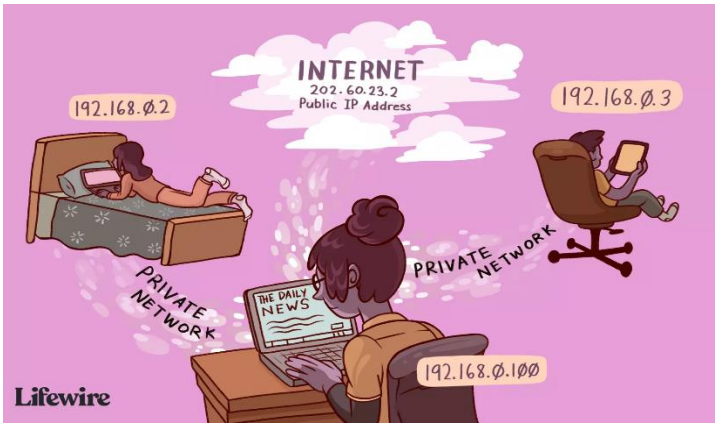
Types of IP Address by Visibility

There are 2 types of IP Addresses- Public, Private.

I. **Public IP Address:** Public is an Internet Protocol address, encrypted by various servers/devices. That's when you connect these devices with your internet connection. This is the same IP address we show on our homepage. Some even call this their external IP address. A public Internet Protocol address is an Internet Protocol address accessed over the Internet. Like the postal address used to deliver mail to your home, the public Internet Protocol address is a different international Internet Protocol address assigned to a computer device. The web server, email server, and any server device that has direct access to the Internet are those who will enter the public Internet Protocol address. Internet Address Protocol is unique worldwide and is only supplied with a unique device.

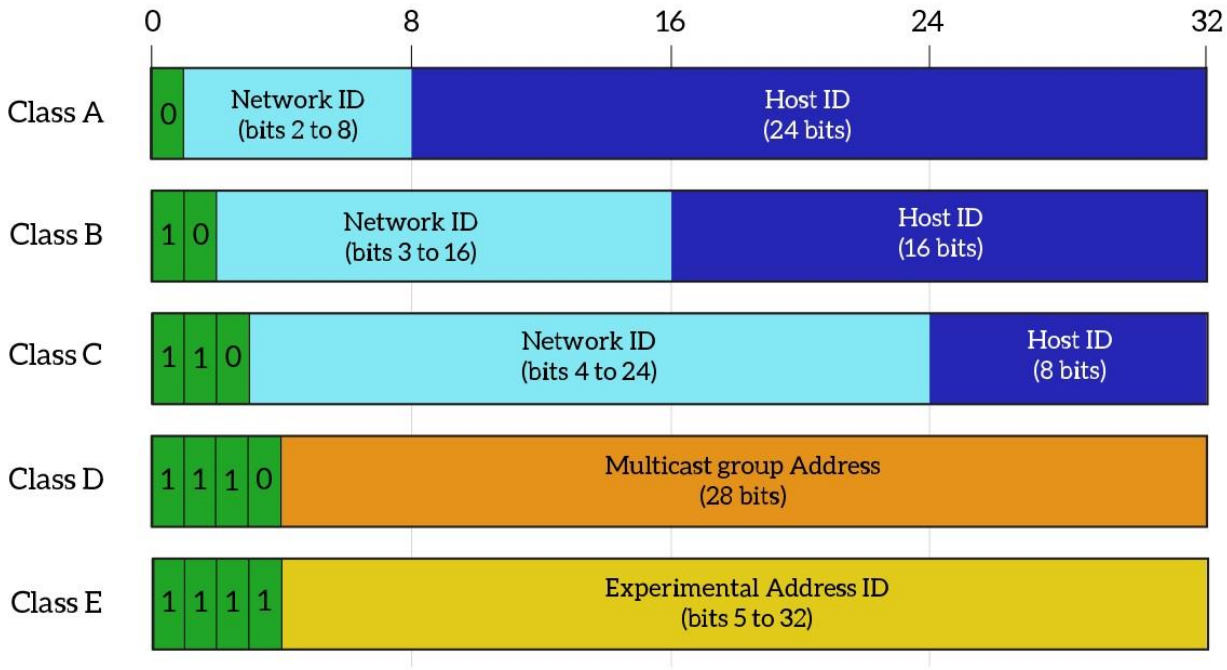


II. **Private IP Address:** Everything that connects to your Internet network has a Private IP. This includes computers, smartphones, tablets and Bluetooth-enabled devices such as speakers, printers, or smart TV. Therefore, Router generates private IP addresses that are unique identifiers for each device that separates the network.



IP Address Class

Internet Protocol defines five classes of IP's: class A, B, C, D, and E. Each class has a range of valid addresses assigned to it. The value of the first octet tells us the class to which it belongs to. First three classes (class A, B and C) are used for normal host & network addresses. The other two classes are used for other purpose. Class A addresses 127.0.0.0 to 127.255.255.255 cannot be used and is reserved for loopback and diagnostic functions.



In IPv4, the IP address is divided into two parts: the network part and the host part. The division between the network and host parts is determined by the subnet mask. The subnet mask is a 32-bit value that consists of consecutive 1 (binary) in the network part followed by consecutive 0 (binary) in the host part. In attendance are five IPv4 address classes (Class A to Class E) designated as follows:

- Class A (1.0.0.0 to 126.255.255.255):

Class A addresses have the first bit of the IP address set to 0 (MSB), and the network part occupies the first octet (8 bits). The remaining three octets (24 bits) represent the host part. Class A addresses were originally intended for large networks and have a vast number of host addresses.

- Class B (128.0.0.0 to 191.255.255.255):

Class B addresses have the first two bits of the IP address set to 10 (MSB), and the network part occupies the first two octets (16 bits). The remaining two octets (16 bits) represent the host part. Class B addresses were designed for medium-sized networks.

- Class C (192.0.0.0 to 223.255.255.255):

Class C addresses have the first three bits of the IP address set to 110 (MSB), and the network part occupies the first three octets (24 bits). The remaining octet (8 bits) represents the host part. Class C addresses were intended for small networks.

- Class D (224.0.0.0 to 239.255.255.255):

Class D addresses have the first four bits of the IP address set to 1110 (MSB). These addresses are used for multicast purposes, where data can be sent to multiple hosts at a time.

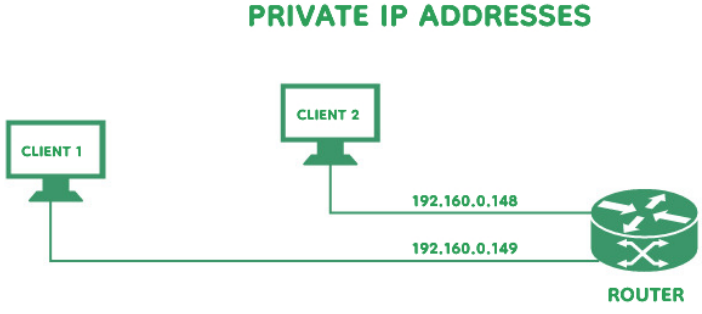
- Class E (240.0.0.0 to 255.255.255.255):

Class E addresses have the first four bits of the IP address set to 1111 (MSB). Class E addresses were reserved for experimental purposes and were not meant for general use.

It's important to note that IP address classification based on classes is mostly a historical concept. In modern networking, subnetting and Classless Inter-Domain Routing (CIDR) are commonly used techniques to allocate IP addresses more efficiently and flexibly regardless of the traditional class boundaries. CIDR allows for the allocation of IP addresses in variable-sized blocks, which enables better address utilization and more efficient routing of the network traffic.

Private IP Address Ranges

Private IP Addresses are those addresses that work within the local network. These addresses are non-routable on the Internet. The address is assigned by the network Roter of your particular device. The unique private IP address is provided to every device which is on the same network.

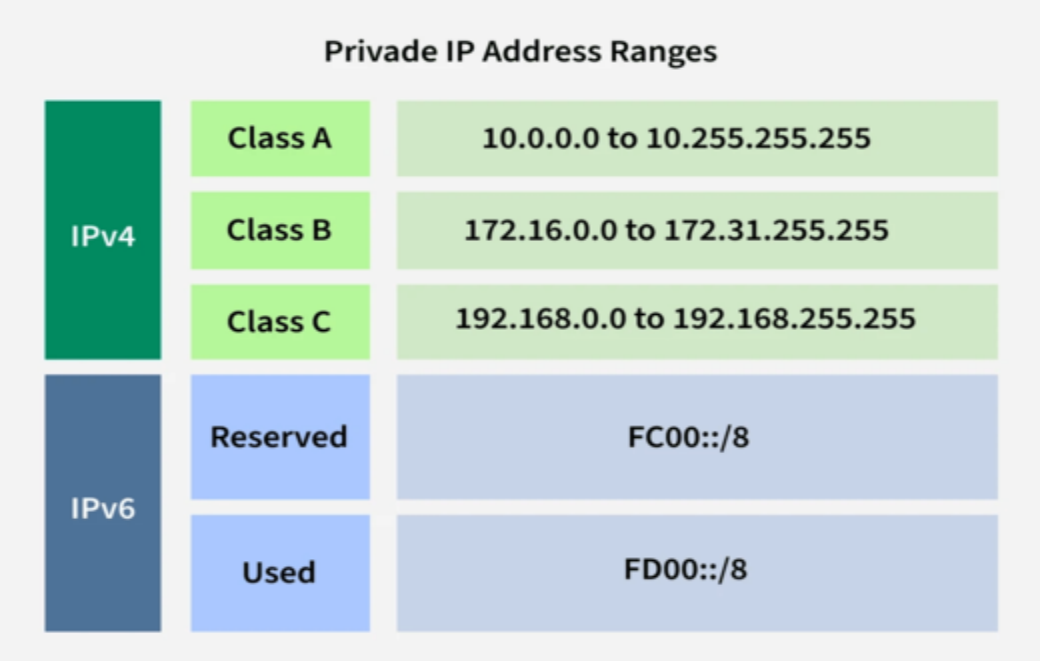


What is a Private IP Address Used for?

- **Home Networks**
- **Business Network**

Private IP Ranges

Private address ranges enable devices on the private network to communicate effectively without conflicting the usage of the public IP addresses, hence the need for it in network management and security.



IPv4 Address Format

An IPv4 address consists of 32 Bit grouped into four Parts of known as octets or bytes. Each octet has 8 bits & these bits can be represented only in 0 or 1 form, and when they grouped together, they form a binary number. Since each octet has 8 bits, it can represent 256 numbers ranging from o to 255. These four octets are represented as decimal numbers, separated by periods known as dotted decimal notation. Example IPv4 address 185.107.80.231 consists of four octets.



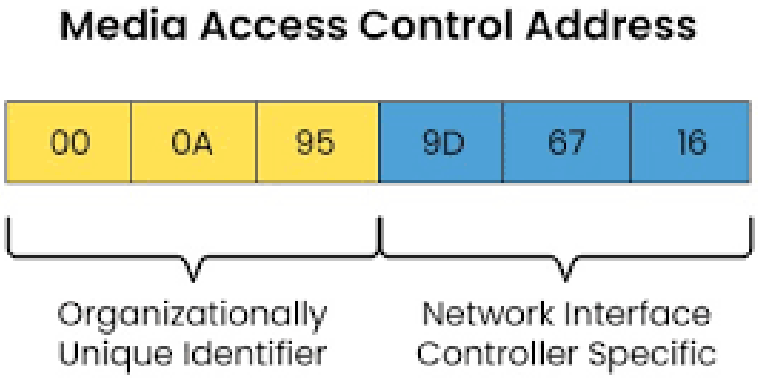
Binary Representation

IPv4 is basically converted into binary form by computer although these are usually seen in decimal form for human readability. Each octet is converted into 8 Bit binary Number. Example **185.107.80.231** in binary looks like

- 185------>10111001
- 107------> 01101011
- 80 ------> 01010000
- 231------> 11100111

MAC Address

MAC Addresses are unique 48-bit hardware numbers of a computer that are embedded into a network Interface card manufacturing. The MAC Address is also known as the physical address in a network device.



Format of MAC Address

To understand what is MAC address is, it is very important that first you understand the format of the MAC Address. A MAC Address is a 12-digit hexadecimal number (48-bit binary number), which is mostly represented by Colon-Hexadecimal notation.

The First 6 digits (say 00:40:96) of the MAC Address identify the manufacturer, called the OUI (Organizational Unique Identifier) or Vendor Information. IEEE Registration Authority Committee assigns these MAC prefixes to its registered vendors. Here are some OUI of well-known manufacturers:

CC:46:D6 - Cisco
3C:5A:B4 - Google, Inc.
3C:D9:2B - Hewlett Packard
00:9A:CD - HUAWEI TECHNOLOGIES CO.,LTD

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